

THARANI RAVI

Data Engineer | Python | ETL

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PROFESSIONAL SUMMARY

Data Engineer with internship experience developing ETL pipelines and data ingestion workflows using Python, SQL, PostgreSQL, Prefect, Docker, and REST APIs. Experienced in transforming macroeconomic datasets into analytics-ready databases, designing normalized relational schemas, and implementing historical backfills, incremental ingestion, data validation, and retry logic. Strong foundation in data pipelines, data modelling, and reliable data processing, with experience working with external financial data APIs.

TECHNICAL SKILLS

Programming: Python, SQL

Data Engineering: ETL, Data Pipelines, Data Ingestion, Data Transformation, Data Validation, Incremental Ingestion, Historical Backfill, Prefect

Databases & Data Modelling: PostgreSQL, Relational Data Modelling, Normalized Database Schemas

Data & APIs: REST APIs, Federal Reserve Economic Data (FRED) API

Libraries: Pandas, NumPy, Plotly

Tools: Docker, Git, GitHub, Jupyter

PROFESSIONAL EXPERIENCE

AI & Data Engineering Intern — Kaar Technologies, Chennai

January 2026 – Present

- Developed Prefect-based ETL pipelines using Python, Prefect, PostgreSQL, and REST APIs to ingest and transform **3+ macroeconomic datasets** into analytics-ready databases. Designed normalized PostgreSQL schemas across **5+ relational tables** supporting economic releases, observations, and metadata.
- Implemented historical backfill and incremental ingestion workflows with validation, retry logic, and reusable pipeline components for reliable data processing. Evaluated external financial market data APIs to understand authentication mechanisms, data models, and ingestion patterns, supporting future ETL pipeline design.

PROJECTS

FRED Macroeconomic Data Pipeline | Python, SQL, PostgreSQL, REST APIs, Prefect, ETL

- Built an end-to-end pipeline for Federal Reserve Economic Data ingestion and transformation, with normalized relational storage and analytics-ready outputs.
- Implemented historical backfill and incremental ingestion patterns with data validation and reusable pipeline components for reliable processing.

Prediction Market Data Pipeline | Python, REST APIs, ETL, Data Transformation

- Developed a data pipeline for prediction-market data ingestion and transformation using API-driven workflows and structured processing patterns.

Smart Traffic Management System | Python, OpenCV, Data Preprocessing

- Pre-processed traffic datasets and engineered features to train a machine learning model for adaptive signal timing.
- Ran simulations to evaluate model-driven traffic control logic and validate signal timing adjustments against vehicle density data.

CERTIFICATIONS

- Anthropic AI Fluency & Claude AI Learning Path – Anthropic | 2026
- Deloitte Data Analytics Job Simulation – Forage | 2026
- Oracle Fusion AI Agent Studio Certified Foundations Associate – Oracle University | 2025
- Accenture Data Analytics and Visualization Job Simulation – Forage | 2024
- Python for Data Science – NPTEL | 2024

PUBLICATIONS

- IEEE Conference Publication — "GPS-Based Power Theft Detection and Alert System Using GSM."

ACHIEVEMENTS

- IEEE YESIST'12 Innovation Challenge — Finalist, Tunisia (2023–24).
- Best Paper Award — INNOVISTA'23, Robotic Surgery Advances.

EDUCATION

B.E., Electronics and Communication Engineering (2022 – 2026) - RMK Engineering College | CGPA: 8.13